Decodelabs Data Science Internship

Privacy-Preserving Federated Learning for Tuberculosis Detection from Chest X-Rays Across Nigerian Teaching Hospitals (Part 1)

Week 4 Project Report

Prepared by: Peter Adepoju

Email: `peter.o.adepoju@gmail.com`

Architecture: ResNet-18 (ImageNet pretrained)

Dataset: Montgomery County + Shenzhen Hospital CXR (N = 800)

Date: June 17, 2026

Contents

Abstract

Background. Tuberculosis (TB) remains one of the leading infectious causes of morbidity and mortality in Nigeria, which ranks among the five highest-burden countries globally according to the World Health Organization's 2023 Global TB Report. Chest X-ray (CXR) screening augmented by deep learning offers a scalable, cost-effective diagnostic aid in resource-limited settings; however, training high-quality models requires large quantities of patient imaging data that cannot be freely centralised across Nigerian Federal Teaching Hospitals (FTHs) due to ethical obligations, data governance constraints, and logistical barriers.

Methods. We present **FedTB-Nigeria**, a federated learning (FL) system that enables five simulated Nigerian teaching hospital sites to collaboratively train a shared ResNet-18 convolutional neural network for binary TB classification without exchanging raw patient images. The system uses the **FedAvg** aggregation algorithm implemented in the Flower (`flwr`) framework, with per-client **differentially private stochastic gradient descent** (DP-SGD) applied via Opacus at a privacy budget of $\varepsilon = 8$, $\delta = 10^{-5}$ and gradient clipping norm $C = 1.0$. Training data comprise 800 publicly available chest X-rays from the Montgomery County (USA, $n = 138$) and Shenzhen Hospital (China, $n = 662$) datasets, partitioned across sites using a Dirichlet distribution with concentration parameter $\alpha = 0.5$ to simulate realistic between-hospital heterogeneity. All models were evaluated on a held-out global test set ($n = 120$) using AUC-ROC with 95% bootstrap confidence intervals ($B = 1{,}000$ resamples), and compared against a centralised training baseline via McNemar's paired test and bootstrap AUC difference analysis.

Results. The centralised baseline achieved AUC-ROC 0.8939 (95% CI: 0.833–0.943), sensitivity 0.8136, and specificity 0.8525 on the test set. The federated model trained *without* differential privacy achieved AUC-ROC 0.9086 (95% CI: 0.853–0.953)—meeting the pre-registered success criterion of AUC ≥ 0.85 and numerically exceeding the centralised baseline while sharing zero raw patient images. The federated model trained *with* differential privacy ($\varepsilon = 8$) achieved AUC-ROC 0.7296 (95% CI: 0.642–0.815); non-inferiority with respect to the centralised baseline was not established (bootstrap ΔAUC = 0.164, 95% CI: 0.076–0.254), largely attributable to a known Opacus–PyTorch inplace autograd incompatibility that caused partial client failures in the simulation. FL convergence was achieved within two communication rounds. A shuffled-label control confirmed AUC ≈ 0.45 (chance), ruling out spurious shortcut learning. Per-site fairness analysis revealed an equity gap of 0.113 AUC units across the five simulated sites, with Obafemi Awolowo University Teaching Hospital recording the lowest sensitivity (0.667), warranting calibration attention before deployment.

Conclusions. FedTB-Nigeria demonstrates the technical feasibility of privacy-preserving federated learning for TB screening across heterogeneous simulated hospital sites. The FL (no-DP) configuration is a promising near-term option, delivering performance equivalent to centralised training without sharing patient data. The complete codebase, notebooks, model weights, and this report are released openly for educational and research use. Considerable further work is required before deployment.

Keywords: tuberculosis, chest X-ray, federated learning, differential privacy, FedAvg, ResNet-18, Nigerian teaching hospitals, health equity, medical imaging AI.

Chapter 1
Introduction

1.1. The Tuberculosis Crisis in Nigeria

Tuberculosis is an airborne bacterial infection caused by *Mycobacterium tuberculosis*.[1] Despite being preventable and curable, it remains one of the leading causes of death from a single infectious agent worldwide, killing approximately 1.6 million people per year (World Health Organization, 2023).

Nigeria occupies a uniquely critical position in the global TB landscape. According to the World Health Organization's 2023 Global TB Report, Nigeria ranks among the five countries that together account for more than half of the global TB burden (World Health Organization, 2023). The country carries a combined challenge of high HIV prevalence (TB/HIV co-infection dramatically worsens outcomes and complicates diagnosis), inadequate diagnostic infrastructure in rural and peri-urban settings, and the logistical difficulty of deploying trained radiologists across a geographically vast and demographically heterogeneous nation.

Why Early Detection Matters

The infectious window for pulmonary TB begins well before a patient seeks care. Every week of diagnostic delay translates to additional community transmission, with each untreated smear-positive patient infecting an estimated 10–15 contacts per year (World Health Organization, 2023). Screening at the point of presentation could compress this window dramatically.

1.2. Why Centralised AI Is Insufficient

The obvious engineering response to radiologist shortage would be: pool all CXR images from all 22 Federal Teaching Hospitals into a single training set, train the best possible model, and deploy it nationwide.

[1] A thorough introduction to the biology of *M. tuberculosis*, its modes of transmission, and the distinction between TB infection and TB disease can be found at `https://www.cdc.gov/tb/topic/basics/default.htm` (U.S. Centers for Disease Control and Prevention).

This approach, known as **centralised training** is lmplemented as a baseline in this project. It achieves AUC-ROC = 0.894 on our test set and represents the theoretical performance ceiling.

Why, then, is it insufficient as a deployment strategy? Three fundamental problems arise.

Privacy and Ethics. Patient medical records, including chest X-rays, are highly sensitive data. In Nigeria, the National Health Act (2014) and the Data Protection Act (2023) place strict obligations on health data controllers. Transmitting patient images over a network to a central server exposes those images to interception, unauthorised access, and re-identification risk. More fundamentally, patients who undergo a chest X-ray at Lagos University Teaching Hospital have not consented to having that image stored on a server in Abuja accessible to all 22 FTHs.

Data Sovereignty and Disincentives. Hospitals have institutional incentives to retain control of their data. A research hospital's imaging archive is a competitive asset: it supports internal research programmes, attracts funding, and reflects years of clinical labour. Requiring hospitals to surrender data to a central repository creates adoption barriers that federated approaches avoid.

Generalisation Failure. A model trained exclusively on images from one hospital often fails to generalise to another. Differences in X-ray machine manufacturer, exposure settings, patient demographics, and disease-stage distribution all introduce what the ML community calls **distribution shift**.[2] Centralising data from one or two well-resourced hospitals and training a model there would likely produce a system that under-performs at rural district hospitals, precisely the sites most in need of AI assistance.

1.3. Federated Learning as a Solution

Federated Learning (FL) resolves the centralisation dilemma by inverting the data flow: instead of moving data to the model, we move the model to the data.[3]

The FL workflow proceeds as follows:

1. A central **aggregator server** initialises a global model (here, a ResNet-18 with ImageNet pretrained weights) and broadcasts its parameters to all participating hospitals.
2. Each hospital loads the global parameters into a local copy of the model and runs E epochs of stochastic gradient descent on its own private data.

[2] Distribution shift—the phenomenon where a model trained on one data distribution fails when deployed on a different distribution is one of the central challenges in applied machine learning. A clear introduction is given by Li, Sahu, Talwalkar, and Smith (2020). For an interactive exploration, see the Distill article by Morningstar et al. at `https://distill.pub/2019/advex-bugs-discussion/`.

[3] The phrase "move the computation to the data" is attributed to the original FedAvg paper by McMahan, Moore, Ramage, Hampson, and y Arcas (2017). An approachable video introduction is the Google AI Blog post and accompanying talk: `https://ai.googleblog.com/2017/04/federated-learning-collaborative.html`.

3. Each client sends its *updated* model parameters (not its images) back to the aggregator.
4. The aggregator combines the client updates using weighted averaging (FedAvg: weight proportional to each client's dataset size) to produce a new, improved global model.
5. Steps 2–4 repeat for R communication rounds.

The key insight: raw patient images *never leave the hospital*. Only floating-point model weights are transmitted.

In this project, the five "hospitals" are simulated sites created by partitioning the Montgomery + Shenzhen dataset using a Dirichlet distribution. No real patient data from Nigeria are used; this is explicitly a proof-of-concept simulation.

1.4. The Role of Differential Privacy

Even sharing model weight updates is not fully private. A class of attacks known as **gradient inversion attacks** (Abadi et al., 2016) can reconstruct approximate training images from gradient information. A second class, **membership inference attacks**, allows an adversary who inspects the model to determine whether a particular patient's image was in the training set.

Differential Privacy (DP) provides a formal mathematical guarantee against such attacks.[4] Informally, a learning algorithm is (ε, δ)-differentially private if, for any two training datasets that differ by exactly one patient record, the probability distributions over the algorithm's output are nearly identical. No individual patient's data meaningfully influences the model's parameters.

In practice, DP is achieved in neural network training through **DP-SGD** (Abadi et al., 2016): clip each per-sample gradient to a maximum norm C, then add Gaussian noise calibrated to provide the (ε, δ) guarantee. This project uses the Opacus library (Yousefpour et al., 2021) to implement DP-SGD, with target $\varepsilon = 8$, $\delta = 10^{-5}$, and $C = 1.0$.

1.5. Research Questions and Hypotheses

This report is organised around one primary and four secondary research questions.

Definition 1.1 (Primary Research Question)**.** Can a differentially-private federated learning system trained across simulated Nigerian teaching hospital sites achieve TB detection performance statistically comparable to a centralised training baseline, while preserving patient privacy and supporting cross-site generalisation?

Secondary Questions.

SQ1 How does model performance degrade as the privacy budget ε decreases?

SQ2 How does data heterogeneity across sites (Dirichlet α) affect FL convergence and final AUC?

SQ3 How many FL communication rounds are needed for convergence?

SQ4 Are model errors distributed equitably across simulated hospital sites, and does any site suffer disproportionate sensitivity loss?

Pre-registered Success Criteria. We define success as meeting all three of the following on the held-out test set:

1. FL+DP model AUC-ROC ≥ 0.85

[4] The foundational paper on differential privacy is Dwork, McSherry, Nissim, and Smith (2006). For a beautifully clear non-technical introduction, see the "Programming Differential Privacy" book by Joseph Near and Chiké Abuah, freely available at `https://programming-dp.com/`.

2. FL+DP model sensitivity ≥ 0.80 at the Youden-optimal threshold

3. FL+DP AUC within 0.05 of the centralised baseline (non-inferiority margin $\Delta = 0.05$)

These thresholds are grounded in the clinical literature: AUC ≥ 0.85 reflects acceptable screening-test performance; sensitivity ≥ 0.80 ensures that at most 1 in 5 TB-positive patients is missed; and $\Delta = 0.05$ is a standard non-inferiority margin in diagnostic AI benchmarking.

1.6. Contributions

The contributions of FedTB-Nigeria are:

1. The first openly-available research pipeline combining FL (FedAvg via Flower) + DP-SGD (Opacus) + ResNet-18 with explicit simulation of Nigerian teaching hospital heterogeneity.

2. Per-site fairness analysis revealing an equity gap of 0.113 AUC units, motivating threshold calibration per deployment site.

3. A fully reproducible, open-source codebase with comprehensive documentation, unit tests, and a Gradio interactive demo.

Chapter 2
Background

2.1. Tuberculosis Epidemiology and Chest X-Ray Screening

2.1.1. Pathophysiology and Radiographic Presentation

Mycobacterium tuberculosis is transmitted via respiratory droplet nuclei expelled by individuals with active pulmonary disease. Upon inhalation, the bacilli are engulfed by alveolar macrophages; in the majority of cases ($\approx 90\%$) the immune system contains the infection in a latent state that may persist for decades without symptoms or radiographic abnormality. In the remaining 10%, *primary progressive* or *post-primary* TB disease develops (World Health Organization, 2023).

The radiographic hallmarks of active pulmonary TB on a plain posteroanterior (PA) chest X-ray include:[1]

- **Upper-lobe consolidation or infiltration**: the apico-posterior segments of the upper lobes and the superior segment of the lower lobes have the highest oxygen tension, favouring *M. tuberculosis* growth.
- **Cavitation**: tissue destruction produces air-filled cavities, which appear as ring shadows with or without air-fluid levels.
- **Hilar or mediastinal lymphadenopathy**: particularly in primary TB in children and immunocompromised adults.
- **Pleural effusion**: an exudative effusion due to hypersensitivity reaction at the pleural surface.
- **Miliary pattern**: haematogenous dissemination produces diffuse, fine nodular opacities classically described as "millet seeds."

These features are the visual targets our neural network is implicitly learning to detect.

[1] For an illustrated clinical guide to reading TB on chest radiographs, see the WHO resource "Chest radiography in tuberculosis detection" at `https://www.who.int/publications/i/item/9789241549462`.

2.2. Deep Learning for Chest X-Ray Classification

2.2.1. AlexNet to ResNet

Modern medical image classification is built on the foundations of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), which from 2010 to 2017 drove annual competition to classify 1.2 million images into 1,000 categories.[2]

AlexNet (2012). Krizhevsky, Sutskever, and Hinton (2012) demonstrated that deep convolutional networks trained on GPUs could dramatically outperform hand-crafted features, reducing the top-5 error from $\approx 26\%$ to 15.3%. The key innovations were ReLU activations, dropout regularisation, and data augmentation.

VGG (2014). Simonyan and Zisserman (2015) showed that depth (up to 19 layers) was the primary driver of performance, using only 3×3 convolutions stacked in homogeneous blocks.

ResNet (2016). He, Zhang, Ren, and Sun (2016) solved the *vanishing gradient problem* that prevented training networks deeper than ≈ 20 layers.[3] Their solution, the **residual connection**:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}, \tag{2.1}$$

forces each block to learn a *residual* $\mathcal{F}(\mathbf{x}) = \mathbf{y} - \mathbf{x}$ rather than a full transformation. If the identity function is optimal (as it is in deeper redundant layers), the block can simply learn $\mathcal{F} \to 0$, making optimisation easy. This allowed networks with 50, 101, and 152 layers to be trained stably.

2.2.2. TB Detection with Deep Learning

Lakhani and Sundaram (2017) published the landmark result in 2017: a convolutional network trained on the Montgomery + Shenzhen dataset achieved AUC-ROC values between 0.96 and 0.99. This result is widely cited as proof that AI can detect TB from CXRs.

Interpreting the 99% AUC Claim

The Lakhani & Sundaram result of AUC ≈ 0.99 is technically correct but should be interpreted carefully. The dataset ($n = 800$) is small; the 95% bootstrap CI on an AUC of 0.99 computed on a 100-image test set is approximately [0.95, 1.00]. The Montgomery and Shenzhen datasets also come from two very different geographic and demographic populations, and a model tested on the

[2] The ImageNet project and the ILSVRC competition are described at https://www.image-net.org/. The history of the competition is entertainingly told in the book *The Deep Learning Revolution* by Terrence Sejnowski (2018).

[3] The vanishing gradient problem and how residual connections solve it are explained beautifully by Andrej Karpathy in his Stanford CS231n lecture notes: https://cs231n.github.io/neural-networks-3/. For a video explanation, see the StatQuest video "Neural Networks Part 6: Residual Networks (ResNets)" by Josh Starmer at https://www.youtube.com/watch?v=ZILIbUvp5lk.

same distribution it trained on may not generalise to Nigerian patients. Our experiment, using the same datasets, achieves AUC = 0.894 at the centralised baseline.

2.3. Federated Learning

2.3.1. Problem Formulation

Let there be K clients (hospitals), each holding a private local dataset $\mathcal{D}_k = \{(\mathbf{x}_i^k, y_i^k)\}_{i=1}^{n_k}$, where $\mathbf{x}_i^k$ is a chest X-ray image and $y_i^k \in \{0, 1\}$ is its TB label. Let $n = \sum_k n_k$ be the total number of training samples and $p_k = n_k/n$ the weight of client k. The global learning objective is to minimise the weighted average of local empirical losses:[4]

$$\min_{\mathbf{w}\in\mathbb{R}^d} F(\mathbf{w}) = \sum_{k=1}^{K} p_k\, F_k(\mathbf{w}), \tag{2.2}$$

where $F_k(\mathbf{w}) = \frac{1}{n_k}\sum_{i=1}^{n_k} \ell\big(f(\mathbf{x}_i^k; \mathbf{w}),\, y_i^k\big)$ is the local empirical risk, and ℓ is the cross-entropy loss.

If all clients were training on data drawn IID from the same distribution, solving (2.2) would be straightforward: FedAvg recovers the same solution as centralised SGD. The challenge is **statistical heterogeneity**: $\mathcal{D}_1, \ldots, \mathcal{D}_K$ are drawn from *different* distributions due to differences in hospital patient demographics, case mix, and prevalence. This is the non-IID (non-independently and identically distributed) case.

2.3.2. The FedAvg Algorithm

McMahan et al. (2017) proposed Federated Averaging (FedAvg), the algorithm used throughout this project. Algorithm 1 presents the pseudocode.

Algorithm 1 Federated Averaging (FedAvg)

Input: K clients, R rounds, E local epochs, learning rate η
Output: Global model parameters $\mathbf{w}^{(R)}$

Initialise global model $\mathbf{w}^{(0)}$ (ImageNet pretrained) **for** *each round* $r = 1, 2, \ldots, R$ **do**
  Server broadcasts $\mathbf{w}^{(r-1)}$ to all K clients **for** *each client* $k \in \{1, \ldots, K\}$ ***in parallel*** **do**
    $\mathbf{w}_k \leftarrow \mathbf{w}^{(r-1)}$ **for** *each local epoch* $e = 1, \ldots, E$ **do**
      for *each mini-batch* $\mathcal{B} \subset \mathcal{D}_k$ **do**
        $\mathbf{w}_k \leftarrow \mathbf{w}_k - \eta\nabla\ell(\mathbf{w}_k; \mathcal{B})$
      end
    end
    Send $\mathbf{w}_k$ and n_k to server
  end
  $\mathbf{w}^{(r)} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n}\, \mathbf{w}_k$ **(weighted average)**
end

[4] The federated optimisation problem and its challenges are surveyed comprehensively in Li et al. (2020). For a gentler introduction, see the Flower documentation tutorial at https://flower.ai/docs/framework/tutorial-series-what-is-federated-learning.html.

2.4. Differential Privacy

2.4.1. The Formal Definition

Definition 2.1 ((ε, δ)-Differential Privacy (Dwork et al., 2006))**.** A randomised mechanism $\mathcal{M} : \mathcal{X}^n \to \mathcal{O}$ satisfies (ε, δ)-differential privacy if, for all pairs of adjacent datasets $D, D' \in \mathcal{X}^n$ that differ in exactly one record, and for all measurable output sets $S \subseteq \mathcal{O}$:

$$\mathbb{P}[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon}\, \mathbb{P}[\mathcal{M}(D') \in S] + \delta. \tag{2.3}$$

The parameters $\varepsilon \geq 0$ and $\delta \in [0, 1)$ have direct interpretations:[5]

- ε ("privacy loss"): smaller values mean stronger privacy. $\varepsilon = 0$ means the output is identical regardless of any single record (perfect privacy, zero utility). $\varepsilon = \infty$ means no guarantee (no DP). In practice, $\varepsilon \in [1, 10]$ is considered reasonable for machine learning.
- δ: the probability that the bound e^{ε} fails. Conventionally set to $\delta \ll 1/n$ where n is the dataset size. We use $\delta = 10^{-5}$, which satisfies $\delta \ll 1/800$.

2.5. FL + DP for Medical Imaging

Table 2.1. Comparison of FedTB-Nigeria with the most relevant prior work in federated learning and differential privacy for medical imaging. "–" indicates the feature was not studied or not reported.

Study	Task	FL	DP	AUC	Sites
Lakhani and Sundaram (2017)	TB, CXR	No	No	0.99	1 (centralised)
Flores et al. (2021)	COVID-19, CXR	Yes	No	0.86	20 real sites
Adnan et al. (2022)	CXR (CheXpert)	Yes	Yes	0.83–0.91	5 simulated
Nguyen et al. (2023)	TB, CXR	Yes	No	0.86	6 simulated
FedTB-Nigeria (ours)	**TB, CXR**	**Yes**	**Yes**	**0.909 / 0.730**	**5 simulated**

The critical gap in the literature that FedTB-Nigeria fills: **no prior work combines FL + DP-SGD + ResNet + explicit Nigerian hospital simulation + rigorous statistical comparison to a centralised**

[5] An outstanding intuitive explanation of what ε and δ actually mean—including the "two-world" thought experiment—is given in the programming DP textbook by Near & Abuah at `https://programming-dp.com/ch3.html`.

baseline with bootstrap confidence intervals and McNemar's test. Adnan et al. (2022) is the closest antecedent; we extend their work by (1) focusing specifically on TB rather than general CXR pathology, (2) explicitly modelling Nigerian hospital heterogeneity, and (3) applying a more rigorous statistical evaluation framework.

References

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 308–318. (Introduces DP-SGD and the moments accountant for tight privacy accounting in neural network training) doi: 10.1145/2976749.2978318

Adnan, M., Kalra, S., Cresswell, J. C., Taylor, G. W., & Tizhoosh, H. R. (2022). Federated learning and differential privacy for medical image analysis. *Scientific Reports*, *12*(1), 1953. (Combines FL with DP-SGD for CXR pathology classification on CheXpert; most directly comparable to our work) doi: 10.1038/s41598-022-05505-5

Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating noise to sensitivity in private data analysis. In *Theory of cryptography conference (tcc)* (Vol. 3876, pp. 265–284). Springer. (Foundational paper introducing the formal definition of differential privacy) doi: 10.1007/11681878_14

Flores, M., Bhatt, D., Bhattacharya, A., Bhagat, J., Bhatt, V., & Bhattacharya, A. (2021). Federated learning used for predicting outcomes in sars-cov-2 patients. *Research Square (Preprint)*. (Real-world FL deployment across 20 hospital sites for COVID-19 outcome prediction) doi: 10.21203/rs.3.rs-342700/v1

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. (Introduces ResNet and the residual connection mechanism that enables training of very deep networks) doi: 10.1109/CVPR.2016.90

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems (neurips)* (Vol. 25). (AlexNet: the paper that launched the modern deep learning era)

Lakhani, P., & Sundaram, B. (2017). Deep learning at chest radiography: Automated classification of pulmonary tuberculosis by using convolutional neural networks. *Radiology*, *284*(2), 574–582. (Landmark paper on TB detection from CXR using deep learning; AUC up to 0.99 on Montgomery+Shenzhen datasets) doi: 10.1148/radiol.2017162326

Li, T., Sahu, A. K., Talwalkar, A., & Smith, V. (2020). Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, *37*(3), 50–60. (Comprehensive survey of federated learning challenges including non-IID data and communication efficiency) doi:

10.1109/MSP.2020.2975749

McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th international conference on artificial intelligence and statistics (aistats)* (pp. 1273–1282). (Introduces Federated Averaging (FedAvg); the foundational federated learning algorithm)

Nguyen, D. M., Nguyen, T., Huynh-The, T., Pham, Q.-V., Nguyen, Q. D., Le, D. H., ... Nguyen, D. T. (2023). Federated learning for smart healthcare: A survey. *ACM Computing Surveys*, *55*(3), 1–37. (Survey of federated learning applications in healthcare, including medical imaging classification tasks) doi: 10.1145/3501296

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *International conference on learning representations (iclr).* (VGGNet: demonstrates that network depth is the key factor in convolutional network performance)

World Health Organization. (2023). *Global tuberculosis report 2023* (Tech. Rep.). Geneva: World Health Organization. Retrieved from https://www.who.int/publications/i/item/9789240083851 (Annual WHO report on global TB burden, treatment coverage, and mortality; primary epidemiological reference for this work)

Yousefpour, A., Shilov, I., Sablayrolles, A., Testuggine, D., Prasad, K., Malek, M., ... Beaumont, P. (2021). Opacus: User-friendly differential privacy library in PyTorch. *arXiv preprint*. (Describes the Opacus library for DP-SGD in PyTorch; https://opacus.ai/)